

Physics 2 formula booklet

By: Ioannis Karras

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Constants and unit conversions

Fundamental unit of charge	$e = 1.60 \cdot 10^{-19} \text{ C}$
Avogadro's number	$N_A = 6.02 \cdot 10^{23} \text{ mol}^{-1}$
Coulomb constant	$k = \frac{1}{4\pi\epsilon_0} = 8.99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$
Permittivity of free space	$\epsilon_0 = \frac{1}{4\pi k} = 8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
Electron mass	$m_e = 9.11 \cdot 10^{-31} \text{ kg}$
Proton mass	$m_p = 1.67 \cdot 10^{-27} \text{ kg}$
Converting from electron volts to joules	$1 \text{ eV} = 1.60 \cdot 10^{-19} \text{ J}$ Since an electron volt is obtained by multiplying the fundamental unit of charge e by 1 V, the number is the same as for e
Dielectric strength of dry air	$E_{\text{max}} \approx 3 \cdot 10^6 \text{ V/m}$
Converting from gauss to tesla	$1 \text{ G} = 10^{-4} \text{ T}$
Magnetic constant (permeability of free space)	$\mu_0 = 4\pi \cdot 10^{-7} \text{ Tm/A} = 4\pi \cdot 10^{-7} \text{ H/m}$
Speed of light	$c = 3.00 \cdot 10^8 \text{ m/s}$
Converting from light years to meters	$1 \text{ ly} = 9.46 \cdot 10^{15} \text{ m}$

Planck's constant	$h = 6.63 \cdot 10^{-34} \text{ J}\cdot\text{s} = 4.14 \cdot 10^{-15} \text{ eV}\cdot\text{s}$
hc , Planck's constant times the speed of light	$hc = 1240 \text{ eV}\cdot\text{nm}$
Boltzmann constant	$k_B = 1.38 \cdot 10^{-23} \text{ J/K}$

Symbols and units

Quantity	Symbol	Unit	Name of unit
Charge	q or Q	C	Coulomb
Electric field	E	N/C	
Electric dipole moment	$\vec{p}$	Cm	
Linear charge density	λ	C/m	
Surface charge density	σ	C/m ²	
Electric flux	ϕ	Nm ² /C	
Voltage/potential/ potential difference	V	V	Volt
Capacitance	C	F	Farad
Electrostatic energy density	u_e	J/m ³	
Current	I	A	Ampere
Resistance	R	Ω	Ohm
Resistivity	ρ	Ωm	
Power	P	W	Watt
Magnetic field	$\vec{B}$	T	Tesla
Magnetic dipole moment	$\vec{\mu}$	Am ²	
Magnetization	$\vec{M}$	A/m	
Magnetic susceptibility	χ_m	Unitless	
Relative permeability	K_m	Unitless	

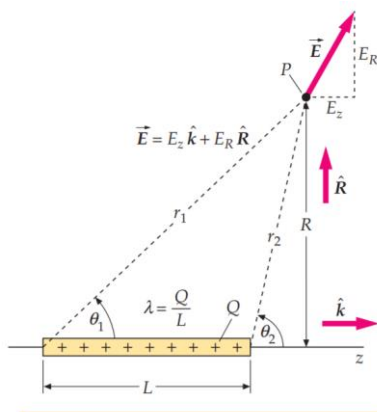
Permeability	μ	Tm/A or H/m (same as permeability of free space)	
Magnetic flux	ϕ_m	Wb	Weber
Inductance	L	H	Henry
Magnetic energy density	u_m	J/m ³	
Inductive reactance	X_L	Ω	Ohm
Capacitive reactance	X_C	Ω	Ohm
Impedance	Z	Ω	Ohm
Frequency	f	Hz	Hertz
Wavelength	λ	m	Meters
Intensity	I	W/m ²	
Refractive index	n	Unitless	

Formulas

Discrete charge distributions (Chapter 21)

Coulomb's Law	$F = \frac{k q_1 q_2 }{r^2}$ where k is the Coulomb constant, $8.99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$ Vector form: $\vec{F}_{12} = \frac{kq_1q_2}{r_{12}^2} \hat{r}_{12}$ where $\hat{r}_{12}$ is the unit vector pointing from the charge q_1 to q_2
Electric field	$\vec{E} = \frac{\vec{F}}{q_0}$ where q_0 is the charge of a small test charge
Coulomb's law for the electric field	$\vec{E} = \frac{kq}{r^2} \hat{r}$ where q is the charge of the field source, r is the distance from the source to the point being considered, and $\hat{r}$ is a unit vector pointing from the source to the point
Electric field due to a system of point charges is additive	$\vec{E}_P = \sum_i \vec{E}_{iP}$ Add the electric fields due to different sources i to obtain the total electric field at point P
Dipole moment	$\vec{p} = q\vec{L}$ where q is the charge at the positive end of the dipole and $\vec{L}$ is the vector going from the negative charge to the positive
Torque on a dipole in an electric field	$\vec{\tau} = \vec{p} \times \vec{E}$ where $\vec{p}$ is the dipole moment and $\vec{E}$ is the electric field
Potential energy of a dipole in an electric field	$E_{pot} = -pE \cos(\theta) = -\vec{p} \cdot \vec{E}$ where $\vec{p}$ is the dipole moment, $\vec{E}$ is the electric field, and θ is the angle between them

Continuous charge distributions (Chapter 22)

Electric field due to a continuous charge distribution	$\vec{E} = \int d\vec{E} = \int \frac{k\hat{r}}{r^2} dq$ where $\hat{r}$ is a unit vector directed from the charge element dq to the field point
Electric field due to a uniformly charged rod on the z-axis	$\vec{E} = E_z \hat{k} + E_R \hat{R}$ where $\hat{k}$ is the z-axis unit vector, $\hat{R}$ is a unit vector in the radial direction, and $E_z = \frac{k\lambda}{R} (\sin(\theta_2) - \sin(\theta_1)) = k\lambda \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ $E_R = -\frac{k\lambda}{R} (\cos(\theta_2) - \cos(\theta_1)) = -k\lambda \left(\frac{\cot(\theta_2)}{r_2} - \frac{\cot(\theta_1)}{r_1} \right)$  FIGURE 22-3 The electric field due to a uniformly charged thin rod. k is the Coulomb constant, $8.99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$
Electric field due to a uniformly charged infinite line	$\vec{E} = \frac{2k\lambda}{R} \hat{R}$ where λ is the linear charge density of the line, R is the radial distance from the line to the point being considered, and $\hat{R}$ is the radial unit vector from the line to the point
Electric field on the axis of a uniformly charged ring	$E_z = \frac{kQz}{(z^2 + a^2)^{3/2}}$ where Q is the charge of the ring, z is the distance from the ring on its axis, and a is the radius of the ring

Electric field on the axis of a uniformly charged disk	$E_z = \text{sign}(z) \cdot 2\pi k\sigma \left(1 - \frac{1}{\sqrt{1 + \frac{R^2}{z^2}}} \right)$ where z is the distance from the disk on its axis, R is the radius of the disk, and σ is its surface charge density
Electric field from a uniform plane of charge on the xy-plane	$E_z = \text{sign}(z) \cdot 2\pi k\sigma = \text{sign}(z) \cdot \frac{\sigma}{2\epsilon_0}$ where ϵ_0 is the permittivity of free space, $8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$, z is the z-axis coordinate of the field point, and σ is the surface charge density of the plane
Electric flux	$\phi = \int_S \vec{E} \cdot \hat{n} dA$ where S is the surface you are calculating the flux through, $\vec{E}$ is the electric field, $\hat{n}$ is a unit vector normal to the surface and pointing outside of it, and dA is an infinitesimal area element If the surface is not curved and the field is uniform, this becomes $\phi = \vec{E} \cdot \hat{n}A = EA \cos(\theta) = E_n A$ where θ is the angle between $\vec{E}$ and $\hat{n}$, A is the area of the surface, and E_n is the component of $\vec{E}$ normal to the surface If $\vec{E}$ is always in the same direction as $\hat{n}$ and the field is constant all over the surface, the equation becomes $\phi = EA$ where E is the magnitude of the electric field and A is the area of the surface
Gauss's Law	$\phi_{\text{net}} = \oint_S \vec{E} \cdot \hat{n} dA = \frac{Q_{\text{inside}}}{\epsilon_0}$ The net outward flux through any closed surface is equal to the net charge inside the surface divided by $\epsilon_0 = 8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
Electric field due to a uniformly charged spherical shell	$E = \frac{kQ}{r^2} \text{ when } r > R$ $E = 0 \text{ when } r < R$ where Q is the charge of the shell, r is the distance from the center of the shell to the point being considered, and R is the radius of the shell. The field is directed away from the center of the shell

Electric field due to a uniformly charged sphere	$E = \frac{kQ}{r^2} \text{ when } r \geq R$ $E = \frac{kQ}{R^3} r \text{ when } r \leq R$ where Q is the charge of the sphere, r is the distance from the center of the sphere to the point being considered, and R is the radius of the sphere. The field is directed away from the center of the sphere
Normal electric field just outside the surface of a conductor	$E_n = \frac{\sigma}{\epsilon_0}$ where σ is the surface charge density and ϵ_0 is the permittivity of free space, $8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$

Electric potential (Chapter 23)

Potential difference	$\Delta V = V_b - V_a = \frac{\Delta E_{\text{pot, e}}}{q_0} = - \int_a^b \vec{E} \cdot d\vec{l}$ where a and b are two points, $\Delta E_{\text{pot, e}}$ is electric potential energy, and $\vec{E}$ is electric field
Relationship between potential energy and potential	$E_{\text{pot, e}} = q_0 V$ where q_0 is a charge in the electric field
Electric potential	$V = \frac{kq}{r}$ where q is the source charge, r is the distance from the source charge, and k is the Coulomb constant, $8.99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$ For a reference 0 potential somewhere other than infinitely far away, $V = \frac{kq}{r} - \frac{kq}{r_{\text{ref}}}$ where r_{ref} is the distance from the reference point to the source point
Electric potential energy of a two-charge system	$E_{\text{pot, e}} = \frac{kq_1 q_2}{r}$
Electric potential due to a system of point charges is additive	$V = \sum_i \frac{kq_i}{r_i}$
Relationship between electric field and potential	$\vec{E} = -\vec{\nabla} V$

Potential due to a continuous charge distribution	$V = \int \frac{k dq}{r}$
Potential on the axis of a charged ring	$V = \frac{kQ}{\sqrt{z^2 + a^2}}$ where Q is the charge of the ring, z is the distance from the ring on its axis, and a is the radius of the ring. $k = 8.99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$
Potential on the axis of a charged disk	$V = 2\pi k\sigma z \left(\sqrt{1 + \frac{R^2}{z^2}} - 1 \right)$ where σ is the surface charge density of the disk, z is the distance from the disk on its axis, and R is the radius of the disk
Potential due to an infinite plane of charge on the yz-plane	$V = V_0 - 2\pi k\sigma x $ where V_0 is the arbitrarily chosen potential at $x = 0$
Potential due to a thin spherical shell	$V = \frac{kQ}{r} \text{ when } r \geq R$ $V = \frac{kQ}{R} \text{ when } r \leq R$ where Q is the charge of the shell, r is the distance of the field point from the center of the shell, and R is the radius of the shell. $k = 8.99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$
Potential due to a uniformly charged sphere	$V = \frac{kQ}{r} \text{ when } r \geq R$ $V = \frac{kQ}{2R} \left(3 - \frac{r^2}{R^2} \right) \text{ when } r \leq R$ where Q is the charge of the sphere, r is the distance of the field point from the center of the sphere, and R is the radius of the sphere
Potential due to a uniformly charged infinite line	$V = 2k\lambda \ln \left(\frac{R_{ref}}{R} \right)$ where λ is the linear charge density, R is the radial distance of the field point, and R_{ref} is the radial distance of the reference point, which cannot be 0 or $\pm\infty$
Electric potential energy of a system of point charges	$E_{\text{pot, e}} = \frac{1}{2} \sum_{i=1}^n q_i V_i$ where q_i is the charge of the ith charge, and V_i is the potential at the location of the ith charge due to the presence of all the other charges in the system

Capacitance (Chapter 24)

Capacitance	$C = \frac{Q}{V}$ where Q is the magnitude of the charge on either plate of a capacitor and V is the potential difference between the plates
Self-capacitance of a spherical conductor	$C = \frac{R}{k} = 4\pi\epsilon_0 R$ where R is the radius of the sphere, $k = 8.99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$, and $\epsilon_0 = 8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
Capacitance of a parallel-plate capacitor	$C = \frac{\epsilon_0 A}{d}$ where A is the area of the surface of each plate that faces the other plate, d is the separation distance between the plates, and $\epsilon_0 = 8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
Capacitance of a cylindrical capacitor	$C = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{R_2}{R_1}\right)}$ where L is the length of the cylinder, R_1 is the radius of the inner cylinder, and R_2 is the radius of the outer cylindrical shell
Energy stored in a capacitor	$E_{\text{pot, e}} = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV = \frac{1}{2} CV^2$
Voltage and electric field across a capacitor	$V = Ed$ where E is the electric field strength between the plates and d is the separation distance between the plates
Electric field in a capacitor in terms of charge	$E = \frac{Q}{\epsilon_0 A} = \frac{\sigma}{\epsilon_0}$ where A is the surface area of each plate, σ is surface charge density, and $\epsilon_0 = 8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
Energy density of an electric field	$u_e = \frac{\text{energy}}{\text{volume}} = \frac{1}{2} \epsilon_0 E^2$ where $\epsilon_0 = 8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
Equivalent capacitance for capacitors in parallel	$C_{eq} = C_1 + C_2 + C_3 + \dots$
Equivalent capacitance for equally charged capacitors in series	$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$

Electric field inside a dielectric	$E = \frac{E_0}{\kappa}$ where E_0 is the electric field without the dielectric and κ is the dielectric constant of the material		
Effect of a dielectric on capacitance	$C = \kappa C_0$ where κ is the dielectric constant of the material and C_0 is the capacitance without the dielectric		
Permittivity of a dielectric	$\epsilon = \kappa \epsilon_0$ where κ is the dielectric constant of the material and ϵ_0 is the permittivity of free space, $8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$		
Dielectric constants and strengths for different materials	Table 24-1 Dielectric Constants and Dielectric Strengths of Various Materials		
	Material	Dielectric Constant κ	Dielectric Strength, kV/mm
	Air	1.00059	3
	Bakelite	4.9	24
	Gasoline	2.0 (70°F)	
	Glass (Pyrex)	5.6	14
	Mica	5.4	10–100
	Neoprene	6.9	12
	Paper	3.7	16
	Paraffin	2.1–2.5	10
	Plexiglas	3.4	40
	Polystyrene	2.55	24
	Porcelain	7	5.7
	Strontium titanate	240	8
	Transformer oil	2.24	12

Electric current (Chapter 25)

Current	$I = \frac{\Delta Q}{\Delta t}$
Relation between current and drift speed	$I = qnAv_d$ where q is the charge of the charge carriers, n is the number density of the carriers, A is the cross sectional area of the wire they are moving through, and v_d is their drift speed
Current density	$\vec{J} = qn\vec{v}_d$ See above

Current in terms of current density	$I = \int_S \vec{J} \cdot d\vec{A} = \int_S \vec{J} \cdot \hat{n} dA$ where $\vec{J}$ is the current density, $\vec{A}$ is an area element vector normal to the surface, and $\hat{n}$ is a unit vector normal to the surface. If $\vec{J}$ is uniform and the surface is flat, this becomes $I = JA \cos(\theta)$ where A is the area of the surface and θ is the angle between $\vec{J}$ and $\hat{n}$
Number density of atoms	$n_a = \frac{\rho_m N_A}{M}$ where ρ_m is the mass density, M is the molar mass, and $N_A = 6.02 \cdot 10^{23} \text{ mol}^{-1}$
Voltage drop for a length of wire	$V = EL$ where E is the electric field in the wire and L is the length of the wire segment
Resistance (Ohm's Law)	$R = \frac{V}{I}$ $V = IR$ where V is voltage drop and I is current
Resistance from resistivity	$R = \rho \frac{L}{A}$ where ρ is the resistivity of the material, L is the length of the wire, and A is its cross-sectional area
Temperature coefficient of resistivity	$\alpha = \frac{(\rho - \rho_0)/\rho_0}{T - T_0}$ where ρ_0 is the resistivity at temperature T_0 and ρ is the resistivity at temperature T
Power delivered to a device or wire segment	$P = IV = I^2 R = \frac{V^2}{R}$
Power supplied by an ideal emf source	$P = I\mathcal{E}$ where $\mathcal{E}$ is the emf of the source
Terminal voltage of a real battery	$V_{\text{battery}} = \mathcal{E} - Ir$ where $\mathcal{E}$ is the emf of the battery, I is the current flowing through the battery, and r is the internal resistance of the battery
Current from a real battery	$I = \frac{\mathcal{E}}{R + r}$ where $\mathcal{E}$ is the emf of the battery, R is the resistance of the circuit, and r is the internal resistance of the battery

Energy stored in a battery	$E_{\text{stored}} = Q\mathcal{E}$ where Q is the total charge stored in the battery and $\mathcal{E}$ is its emf
Equivalent resistance for resistors in series	$R_{eq} = R_1 + R_2 + R_3 + \dots$
Equivalent resistance for resistors in parallel	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$
Kirchhoff's first rule (the loop rule)	When any closed loop is traversed, the algebraic sum of the changes in potential around the loop must equal 0 $\sum V = 0$
Kirchhoff's second rule (the junction rule)	At any junction in a circuit where the current can divide, the sum of the currents into the junction must equal the sum of the currents out of the junction $\sum I_{\text{in}} = \sum I_{\text{out}}$
Charge when discharging a capacitor in an RC circuit	$Q(t) = Q_0 e^{-t/(RC)} = Q_0 e^{-t/\tau}$ where Q_0 is the initial charge and $\tau = RC$ is the time constant
Current when discharging a capacitor in an RC circuit	$I(t) = I_0 e^{-t/\tau}$ where $\tau = RC$ and I_0 is the initial current, given by $I_0 = \frac{V_0}{R} = \frac{Q_0}{RC}$ where V_0 is the initial voltage across the capacitor and Q_0 is its initial charge
Charge when charging a capacitor in an RC circuit	$Q(t) = C\mathcal{E}(1 - e^{-t/(RC)}) = Q_f(1 - e^{-t/\tau})$ where $\mathcal{E}$ is emf, $Q_f = C\mathcal{E}$ is the final charge, and $\tau = RC$
Current when charging a capacitor in an RC circuit	$I(t) = \frac{\mathcal{E}}{R} e^{-t/(RC)} = I_0 e^{-t/\tau}$ where $I_0 = \frac{\mathcal{E}}{R}$ is the initial current and $\tau = RC$
Joule heating when charging a capacitor in an RC circuit	$W_R = \frac{1}{2} Q_f \mathcal{E}$ where $Q_f = C\mathcal{E}$ is the final charge. The amount of Joule heating is equal to the amount of energy stored in the capacitor, and half the energy $W = Q_f \mathcal{E}$ provided by the source of emf

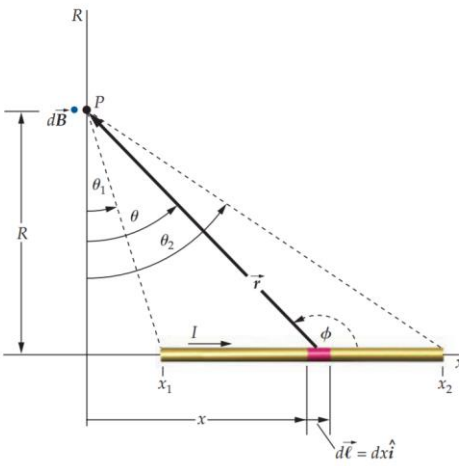
Magnetic fields (Chapter 26)

Superposition of magnetic fields	$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 + \dots$
Magnetic force on a moving charged particle	$\vec{F} = q\vec{v} \times \vec{B}$ $F = qvB \sin(\theta)$ where q is the charge of the particle, $\vec{v}$ is its velocity vector, and $\vec{B}$ is the magnetic field vector
Magnetic force on a straight segment of current-carrying wire	$\vec{F} = I\vec{L} \times \vec{B}$ $F = ILB \sin(\theta)$ where I is the current, $\vec{B}$ is the magnetic field vector, and $\vec{L}$ is a vector whose magnitude is the length of the segment and whose direction is the same as that of the current. For a small current element, this becomes $d\vec{F} = I d\vec{l} \times \vec{B}$
Orbit radius of a charged particle in a magnetic field	$r = \frac{mv}{qB}$ where m is the mass of the particle, v is its speed, q is its charge, and B is the magnetic field strength
Cyclotron period, frequency, and angular speed (charged particle in a circular orbit in a magnetic field)	$T = \frac{2\pi m}{qB}$ $f = \frac{1}{T} = \frac{qB}{2\pi m}$ $\omega = 2\pi f = \frac{q}{m} B$ where m is the mass of the particle, q is its charge, and B is the magnitude of the magnetic field vector
Speed selected by a velocity selector	$v = \frac{E}{B}$
Magnetic dipole moment of a current loop	$\vec{\mu} = NIA\hat{n}$ where N is the number of turns in the loop, I is the current in the loop, A is the area of the loop, and $\hat{n}$ is the unit vector normal to the loop, in the direction given by using the “grip” right hand rule with the looping current direction
Torque on a current loop	$\vec{\tau} = \vec{\mu} \times \vec{B}$ $\tau = \mu B \sin(\theta)$
Potential energy of a magnetic dipole	$E_{\text{pot}} = -\mu B \cos(\theta) = -\vec{\mu} \cdot \vec{B}$

Hall voltage	$V_H = E_H w = v_d B w$ where E_H is the electric field in the Hall effect, w is the width of the conducting strip, v_d is the electron drift speed, and B is the strength of the magnetic field
Number density of electrons in the Hall effect	$n = \frac{ I }{A q v_d} = \frac{ I }{wte v_d} = \frac{ I B}{teV_H}$ where A is the cross-sectional area of the conductor, w is its width, t is its thickness, v_d is the electron drift speed, B is the magnetic field strength, V_H is the Hall voltage, and $e = 1.60 \cdot 10^{-19} \text{ C}$

Sources of the magnetic field (Chapter 27)

Magnetic field of a moving point charge	$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}$ where $\hat{r}$ is the unit vector pointing from the charge to the field point, r is the distance between the charge and the field point, and $\mu_0 = 4\pi \cdot 10^{-7} \text{ Tm/A}$
Magnetic field element produced by a current element (Biot-Savart law)	$d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \vec{r}}{r^2}$ where $Id\vec{l}$ is a current element
Magnetic field on the axis of a current loop	$B_z = \frac{\mu_0}{2} \frac{R^2 I}{(z^2 + R^2)^{\frac{3}{2}}}$ where R is the radius of the loop and z is the distance from the center of the loop. At the center of the loop, this simplifies to: $B = \frac{\mu_0 I}{2R}$ Far from the center of the loop, the equation approximates to: $B_z = \frac{\mu_0}{4\pi} \frac{2\mu}{ z ^3}$ where $\mu = I\pi R^2$ is the magnitude of the magnetic moment of the loop

Magnetic field inside a solenoid	$B_z = \frac{1}{2} \mu_0 n I \left(\frac{z - z_1}{\sqrt{(z - z_1)^2 + R^2}} - \frac{z - z_2}{\sqrt{(z - z_2)^2 + R^2}} \right)$ where n is the number of turns per unit length. The solenoid is assumed to be on the z-axis. z is the z-coordinate of the field point, and z_1 and z_2 are the z-coordinates of the two ends of the solenoid. Far from the ends and inside the solenoid, this approximates to: $B_z = \mu_0 n I$ Again, n is the density of coils per unit length. $\mu_0 = 4\pi \cdot 10^{-7} \text{ Tm/A}$
Magnetic field due to a straight wire segment	$B = \frac{\mu_0 I}{4\pi R} (\sin(\theta_2) - \sin(\theta_1))$  For an infinitely long wire, this simplifies to: $B = \frac{\mu_0 I}{2\pi R}$
Force between two current-carrying wires of equal length	$F = \frac{\mu_0 I_1 I_2}{2\pi R} L$ where R is the distance between the wires and L is their length. $\mu_0 = 4\pi \cdot 10^{-7} \text{ Tm/A}$
Gauss's law for magnetism	$\phi_{\text{m net}} = \oint_S \vec{B} \cdot \hat{n} dA = \oint_S B_n dA = 0$ The net flux of a magnetic field through any closed surface is 0. B_n is the component of $\vec{B}$ normal to the surface at the area element dA
Ampere's law	$\oint_C B_t dl = \oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_C$ where C is a closed curve, and I_C is the current passing through the surface bounded by C. The positive tangential direction for the integral is found using the "grip" right hand rule with the current direction

Magnetic field due to a long, straight wire	$B = \begin{cases} \frac{\mu_0 I}{2\pi R^2} r & \text{if } r \leq R \\ \frac{\mu_0 I}{2\pi r} & \text{if } r \geq R \end{cases}$ where R is the radius of the wire and r is the radial distance from the center of the wire to the field point
Magnetization	$\vec{M} = \frac{d\vec{\mu}}{dV}$ Magnetization is the net magnetic dipole moment per unit volume of a material
Magnetic field inside a magnetized cylinder and far from its ends	$B_m = \mu_0 M$ where M is magnetization
Magnetization in a magnetic field for paramagnetic and diamagnetic materials	$\vec{M} = \chi_m \frac{\vec{B}_{app}}{\mu_0}$ where $\vec{B}_{app}$ is the magnetic field applied on the material and χ_m is the magnetic susceptibility of the material. $\mu_0 = 4\pi \cdot 10^{-7} \text{ Tm/A}$
Magnetic field inside a solenoid magnetizing a material	$\vec{B} = \vec{B}_{app} + \vec{B}_m = \vec{B}_{app} + \mu_0 \vec{M} = \vec{B}_{app}(1 + \chi_m) = \vec{B}_{app} K_m$ where $\vec{B}_{app}$ is the magnetic field applied on the material and $K_m = 1 + \chi_m$ is the relative permeability
Saturation magnetization of a material	$M_s = n\mu$ where n is the number of atoms per unit volume and μ is the magnetic moment of each atom
Permeability	$\mu = (1 + \chi_m)\mu_0 = K_m\mu_0$ where χ_m is magnetic susceptibility and K_m is relative permeability

Induction (Chapter 28)

Magnetic flux	$\phi_m = \int_S \vec{B} \cdot \hat{n} dA = \int_S B_n dA$ where $\vec{B}$ is the magnetic field vector, $\hat{n}$ is a unit normal vector to the surface S, A is an area element, and B_n is the magnetic field component normal to the surface If the surface is flat and the magnetic field is uniform, this simplifies to $\phi_m = \vec{B} \cdot \hat{n} A = BA \cos(\theta) = B_n A$ where A is the area of the surface and θ is the angle between $\vec{B}$ and $\hat{n}$
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Magnetic flux through a coil in a uniform magnetic field	$\phi_m = NBA \cos(\theta)$ where N is the number of turns, B is the magnetic field strength, A is the surface bounded by a loop of the coil, and θ is the angle between the magnetic field and the axis of the coil
Faraday's Law	$\mathcal{E} = -\frac{d\phi_m}{dt}$ EMF is the negative rate of change of magnetic flux
Induced EMF for a stationary circuit in a changing magnetic field	$\mathcal{E} = \oint_C \vec{E}_{nc} \cdot d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \cdot \hat{n} dA = -\frac{d\phi_m}{dt}$ where C is a closed path going around the wire, $\vec{E}_{nc}$ is the nonconservative electric field in the wire, and everything else is as above. The positive tangential direction for the first integral is found by using the “grip” right-hand rule with $\hat{n}$
Magnitude of EMF for a rod moving perpendicular to both the length of the rod and $\vec{B}$	$ \mathcal{E} = Blv$ where B is the magnetic field strength, l is the length of the rod, and v is its speed
Inductance of a long solenoid	$L = \frac{\phi_m}{I} = \mu_0 n^2 Al = \mu_0 \frac{N^2}{l} A$ where n is the number of turns per unit length, A is the area per turn, l is the length of the coil, N is the number of turns, and $\mu_0 = 4\pi \cdot 10^{-7} \text{ Tm/A} = 4\pi \cdot 10^{-7} \text{ H/m}$
Self-induced EMF	$\mathcal{E} = -\frac{d\phi_m}{dt} = -L \frac{dI}{dt}$
Potential difference across an inductor	$\Delta V = \mathcal{E} - Ir = -L \frac{dI}{dt} - Ir$ where r is the internal resistance of the inductor
Mutual inductance	$\phi_{m12} = MI_1$ where ϕ_{m12} is the flux of the magnetic field from circuit 1, through circuit 2, M_{12} is the mutual inductance of the two circuits, and I_1 is the current through circuit 1. Similarly, you can write $\phi_{m21} = MI_2$ with the same mutual inductance value
Mutual inductance for two coaxial solenoids	$M = \frac{\phi_{m12}}{I_1} = \mu_0 n_2 n_1 l \pi r_1^2$ where n_1 and n_2 are the number of turns per unit length for the inner and outer solenoids respectively, l is the length of the solenoids, r_1 is the radius of the inner solenoid, and $\mu_0 = 4\pi \cdot 10^{-7} \text{ Tm/A}$

Energy stored in an inductor	$E_{\text{pot, m}} = \frac{1}{2}LI^2$
Magnetic energy density	$u_m = \frac{B^2}{2\mu_0}$
Current when energizing an inductor in an RL circuit	$I(t) = \frac{\mathcal{E}_0}{R}(1 - e^{-(R/L)t}) = I_f(1 - e^{-t/\tau})$ where $I_f = \frac{\mathcal{E}_0}{R}$ is final current and $\tau = \frac{L}{R}$ is the time constant. $\mathcal{E}_0$ is the emf of the battery
Current when deenergizing an inductor in an RL circuit	$I(t) = I_0 e^{-t/\tau}$ where I_0 is the initial current, which is the same as the final current when the inductor was being energized. τ is the same time constant as above

AC circuits (Chapter 29)

EMF, current, and power from an AC generator	$\mathcal{E} = \mathcal{E}_{\text{peak}} \cos(\omega t)$ $I = I_{\text{peak}} \cos(\omega t)$ $P = I_{\text{peak}}^2 R \cos^2(\omega t)$ $P_{\text{av}} = \frac{1}{2} I_{\text{peak}}^2 R$ ω is the angular speed of the coil R is the total resistance of the circuit $\mathcal{E}_{\text{peak}}$ is the peak EMF produced by the generator $I_{\text{peak}} = \frac{\mathcal{E}_{\text{peak}}}{R}$ P_{av} is average power
RMS (root-mean-square) current and EMF	$I_{\text{rms}} = \sqrt{(I^2)_{\text{av}}}$ $\mathcal{E}_{\text{rms}} = \sqrt{(\mathcal{E}^2)_{\text{av}}}$ where $(I^2)_{\text{av}}$ is the average of the squared current and $(\mathcal{E}^2)_{\text{av}}$ is the average of the squared EMF
RMS current and EMF in terms of the peak value	$I_{\text{rms}} = \frac{1}{\sqrt{2}} I_{\text{peak}} \approx 0.707 I_{\text{peak}}$ $\mathcal{E}_{\text{rms}} = \frac{1}{\sqrt{2}} \mathcal{E}_{\text{peak}} \approx 0.707 \mathcal{E}_{\text{peak}}$ Generally, the rms value of both cos and sin is $1/\sqrt{2}$, so the rms value of anything that varies sinusoidally with time is the peak value divided by $\sqrt{2}$

Average power from RMS current and EMF	$P_{av} = (I_{rms})^2 R = \mathcal{E}_{rms} I_{rms}$ where R is the total resistance of the circuit
RMS current from RMS voltage	$I_{rms} = \frac{V_{rms}}{R}$
Inductive reactance	$X_L = \omega L = 2\pi f L$ where ω is the angular frequency of the current, f is its frequency, and L is inductance
Peak and RMS current in an inductor supplied with AC current	$I_{peak} = \frac{V_{L peak}}{X_L}$ $I_{rms} = \frac{V_{L rms}}{X_L}$ where V_L is the voltage drop across the inductor and X_L is its inductive reactance
Capacitive reactance	$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$ where ω is the angular frequency of the current, f is its frequency, and C is capacitance
Peak and RMS current in a capacitor supplied with AC current	$I_{peak} = \frac{V_{C peak}}{X_C}$ $I_{rms} = \frac{V_{C rms}}{X_C}$ where V_C is the voltage drop across the capacitor and X_C is its capacitive reactance
Current in components in an AC circuit by themselves	If the EMF is given by $V = V_{peak} \cos(\omega t)$ then the current of either a resistor, capacitor, or inductor alone is $I_R = \frac{V}{R} = I_{peak} \cos(\omega t)$ $I_C = \frac{V}{X_C} = I_{peak} \cos\left(\omega t + \frac{\pi}{2}\right)$ $I_L = \frac{V}{X_L} = I_{peak} \cos\left(\omega t - \frac{\pi}{2}\right)$ where X_L and X_C are the inductive and capacitive reactances
Turns and voltage in a transformer	$\frac{N_2}{N_1} = \frac{V_2}{V_1}$
Voltage and current in a transformer	$V_{1 rms} I_{1 rms} = V_{2 rms} I_{2 rms}$

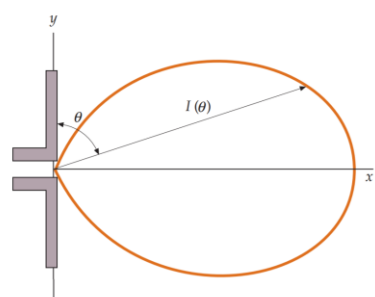
LC circuits	$\omega = 2\pi f = \frac{1}{\sqrt{LC}}$ $Q = Q_{\text{peak}} \cos(\omega t)$ $I = -I_{\text{peak}} \sin(\omega t)$ $I_{\text{peak}} = \omega Q_{\text{peak}}$ $E_{\text{total}} = E_{\text{pot, e}} + E_{\text{pot, m}} = \frac{1}{2} \frac{Q_{\text{peak}}^2}{C}$
Series RLC circuits (both driven and undriven)	$I = I_{\text{peak}} \cos(\omega t - \delta)$ $\tan(\delta) = \frac{X_L - X_C}{R}$ $Z = \sqrt{R^2 + (X_L - X_C)^2}$ $I_{\text{peak}} = \frac{V_{\text{app peak}}}{Z}$ $I = \frac{V_{\text{app peak}}}{Z} \cos(\omega t - \delta)$ δ is the phase constant/angle X_L and X_C are the inductive and capacitive reactances Z is the impedance V_{app} is the potential drop across the entire RLC combination, equal to the EMF of the generator

Driven series RLC circuits	Condition for resonance: $X_L = X_C$ which also leads to $Z = R$ Resonance angular frequency: $\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$ Average power delivered: $P_{av} = \frac{1}{2} I_{peak}^2 R = (I_{rms})^2 R$ $P_{av} = \frac{1}{2} V_{app\ peak} I_{peak} \cos(\delta) = V_{app\ rms} I_{rms} \cos(\delta)$ Power factor: $P_{factor} = \cos(\delta)$ Q factor: $Q_{factor} = \frac{\omega_0 L}{R}$ For narrow resonance curves (Q factors above 2 or 3), $Q_{factor} = \frac{\omega_0}{\Delta\omega} = \frac{f_0}{\Delta f}$ δ is the phase constant/angle X_L and X_C are the inductive and capacitive reactances V_{app} is the potential drop across the entire RLC combination, equal to the EMF of the generator
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Maxwell and electromagnetic waves (Chapter 30)

Displacement current	$I_d = \epsilon_0 \frac{d\phi_e}{dt}$ where ϕ_e is the flux of the electric field through a surface bounded by the curve C in Ampere's Law, and $\epsilon_0 = 8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
Generalized form of Ampere's law	$\oint_C \vec{B} \cdot d\vec{l} = \mu_0(I + I_d) = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_e}{dt}$

Maxwell's equations	<u>Gauss's law:</u> $\oint_S E_n dA = \frac{1}{\epsilon_0} Q_{\text{inside}}$ <u>Gauss's law for magnetism:</u> $\oint_S B_n dA = 0$ Alternate form of <u>Faraday's law:</u> $\oint_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int_S B_n dA = -\int_S \frac{\partial B_n}{\partial t} dA$ Alternate form of <u>generalized form</u> of <u>Ampere's law:</u> $\oint_C \vec{B} \cdot d\vec{l} = \mu_0(I + I_d) = \mu_0 I + \mu_0 \epsilon_0 \int_S \frac{\partial E_n}{\partial t} dA$
Relation between magnitudes in an electromagnetic wave	$E = cB$ where E is the magnitude of the electric field, B is the magnitude of the magnetic field, and $c = 3.00 \cdot 10^8$ m/s
Direction of propagation of an electromagnetic wave	This is the direction given by $\vec{E} \times \vec{B}$ Also, if you have an expression for the electric or magnetic field like $\vec{E}(y, t) = E_0 \sin(ky + \omega t) \hat{k}$ you can find the direction of propagation by looking at the argument of the trigonometric function (either cos or sin). First look at the coordinate variable in the first term of the argument. Then, if the term with ω is added, the wave is moving in the negative direction of the coordinate variable. If the term with ω is subtracted, then it is in the positive direction of that variable. Here, since the first term of the argument has the coordinate variable y and the term with ω is added, the direction of propagation is the negative y direction
The speed of electromagnetic waves	$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$
Frequency and wavelength of an electromagnetic wave	$f\lambda = c$ where f is frequency, λ is wavelength, and $c = 3.00 \cdot 10^8$ m/s

Intensity of radiation from a dipole antenna	$I \propto \frac{\sin^2(\theta)}{d^2}$ $I = k \frac{\sin^2(\theta)}{d^2}$ where θ is the angle between the axis of the antenna and the line going from the center of the antenna to the point (see the figure), and d is the distance between the center of the antenna and the point. $\propto$ means “is proportional to” and k is a constant of proportionality 
Energy density in an electromagnetic wave	$u = u_e + u_m = \epsilon_0 E^2 = \frac{B^2}{\mu_0} = \frac{EB}{\mu_0 c}$ where $\epsilon_0 = 8.85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$, $\mu_0 = 4\pi \cdot 10^{-7} \text{ H/m}$, and $c = 3.00 \cdot 10^8 \text{ m/s}$. E and B are the instantaneous electric and magnetic field, respectively
Poynting vector	$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$ Because of the $\vec{E} \times \vec{B}$, this vector is always in the direction of propagation of the wave
Intensity and average energy density of an electromagnetic wave	$I = u_{\text{av}} c = \frac{E_{\text{rms}} B_{\text{rms}}}{\mu_0} = \frac{1}{2} \frac{E_{\text{peak}} B_{\text{peak}}}{\mu_0} = \frac{E_{\text{rms}}^2}{\mu_0 c} = \frac{B_{\text{rms}}^2 c}{\mu_0} = \vec{S} _{\text{av}}$ where u_{av} is average energy density, E is electric field, B is magnetic field, and $\vec{S}$ is the Poynting vector defined above
Intensity in general	$I = \frac{P}{A}$ where P is power and A is the area that power is distributed over
Momentum and energy in an electromagnetic wave	$p_{\text{wave}} = \frac{E_{\text{wave}}}{c}$ where p is momentum, E_{wave} is the wave energy, and $c = 3.00 \cdot 10^8 \text{ m/s}$
Radiation pressure	$P_r = \frac{I}{c} = \frac{E_{\text{peak}} B_{\text{peak}}}{2\mu_0 c} = \frac{E_{\text{rms}} B_{\text{rms}}}{\mu_0 c} = \frac{E_{\text{peak}}^2}{2\mu_0 c^2} = \frac{B_{\text{peak}}^2}{2\mu_0}$ where E is electric field and B is magnetic field

Properties of light (Chapter 31)

Index of refraction/refractive index	$n = \frac{c}{v}$ where c is the speed of light in vacuum, $3.00 \cdot 10^8$ m/s, and v is the speed of light in the medium
Law of reflection	$\theta'_1 = \theta_1$ The angle of reflection is equal to the angle of incidence
Law of refraction (Snell's law)	$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$ where the ns are the refractive indices of the two media. The angles are between the light ray and the normal. Light is refracted towards the normal when entering a medium with a higher refractive index and vice versa
Wavelength of light in a medium	$\lambda_n = \frac{\lambda}{n}$ where λ is the wavelength of light in vacuum and n is the refractive index of the medium
Reflected intensity for normal incidence	$I = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2 I_0$ where I_0 is the incident intensity and the ns are the refractive indices of the two media
Critical angle	$\sin(\theta_c) = \frac{n_2}{n_1}$ where n_1 is the refractive index of the medium the incident light is traveling through and n_2 is the refractive index of the medium it is approaching. This is the angle of incidence for which the angle of refraction is 90°. For angles of incidence greater than the critical angle, total internal reflection occurs. All of this can only happen when the incident light is in a medium with a higher refractive index than the one it is approaching
Malus's law for polarizers	For polarized incident light, $I = I_0 \cos^2(\theta)$ where I is the intensity of the transmitted beam, I_0 is the intensity of the incident beam, and θ is the angle between the plane of polarization of the incident light and the transmission axis of the polarizer For unpolarized incident light, this becomes $I = \frac{1}{2} I_0$

Polarizing angle	$\tan(\theta_p) = \frac{n_2}{n_1}$ where n_1 is the refractive index of the medium the incident light is traveling through and n_2 is the refractive index of the medium it is approaching. This is the angle of incidence at which the reflected and refracted rays are perpendicular to each other. This also means that the angle of reflection/incidence and the angle of refraction add to 90°
Photon energy	$E = hf = \frac{hc}{\lambda}$ where f is frequency, λ is wavelength, $h = 6.63 \cdot 10^{-34}$ J·s and $c = 3.00 \cdot 10^8$ m/s. hc is also used as a combined constant $hc = 1240$ eV·nm

Interference and diffraction (Chapter 33)

Phase difference due to a path-length difference	$\delta = \frac{\Delta r}{\lambda} 2\pi = \frac{\Delta r}{\lambda} 360^\circ$ where Δr is path-length difference and λ is the wavelength of light while it is going through the medium where the path-length difference is located
Phase difference due to reflection	If light traveling in one medium strikes the surface of a medium in which light travels more slowly, there is a 180° phase change in the reflected light If it instead strikes the surface of a medium in which light travels more quickly, there is no phase change in the reflected light
Phase difference in thin films	If one ray undergoes a 180° phase change when reflecting, $\delta = \frac{2t}{\lambda'} 360^\circ + 180^\circ = \frac{2t}{\lambda'} 2\pi + \pi$ If both rays undergo 180° phase changes when reflecting, $\delta = \frac{2t}{\lambda'} 360^\circ = \frac{2t}{\lambda'} 2\pi$ where t is the thickness of the film and λ' is the wavelength of light in the medium of the film
Maxima and minima in two-slit interference	Interference maxima are located at angles θ_m given by $d \sin(\theta_m) = m\lambda, \quad m = 0, 1, 2, \dots$ Interference minima are located at angles θ_m given by $d \sin(\theta_m) = \left(m - \frac{1}{2}\right)\lambda, \quad m = 1, 2, 3, \dots$ where d is the distance between the slits and m is the order number

Distance from the center to the m th bright fringe in two-slit interference	$y_m = m \frac{\lambda L}{d}$ where m is the order number, L is the distance from the slits to the screen, and d is the distance between the slits
Intensity in terms of phase difference in two-slit interference	$I = 4I_0 \cos^2\left(\frac{1}{2}\delta\right)$ where I_0 is the intensity of the light reaching the screen from either slit separately and δ is phase difference, in this case given by $\delta = \frac{d \sin(\theta)}{\lambda} 2\pi$
Points of zero intensity (minima) for a single-slit diffraction pattern	$a \sin(\theta_m) = m\lambda, \quad m = 1, 2, 3, \dots$ where a is the slit width and m is the order number
Relation between angle and distance from the center for the first diffraction minimum in single-slit diffraction	$\tan(\theta_1) = \frac{y_1}{L}$ θ_1 is the angle of the first diffraction minimum y_1 is the distance from the center to this minimum, or half the width of the central maximum L is the distance from the slit to the screen
Number of bright fringes in the central maximum in combined interference and diffraction	$N = 2m - 1$ where m is the order number of the first interference maximum that coincides with the first diffraction minimum, and thus is not visible. This can be found by setting the sin of an interference maximum angle to the sin of the first diffraction minimum angle: $\frac{m\lambda}{d} = \frac{\lambda}{a}$
Rayleigh's criterion for resolution with a circular aperture	$\alpha_c = 1.22 \frac{\lambda}{D}$ where α_c is the critical angular separation and D is the diameter of the aperture
Interference maxima with a diffraction grating	$d \sin(\theta_m) = m\lambda, \quad m = 0, 1, 2, \dots$ Same equation as for the maxima in two-slit interference. d is the distance between slits and m is the order number
Resolving power of a diffraction grating	$R = \frac{\lambda}{ \Delta\lambda } = mN$ $\Delta\lambda$ is the smallest observable difference between two nearby wavelengths, each approximately equal to λ, that may be resolved m is the order number N is the number of illuminated slits